

I225E Statistical Signal Processing

11. Bayesian Estimation

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Estimation Categories

Optimality Criterion:

- A) Classical estimation θ is nonrandom
 - Minimum variance unbiased estimation (MVUE)
 - Maximum likelihood estimation (MLE)
 - Least squared estimation (LSE)

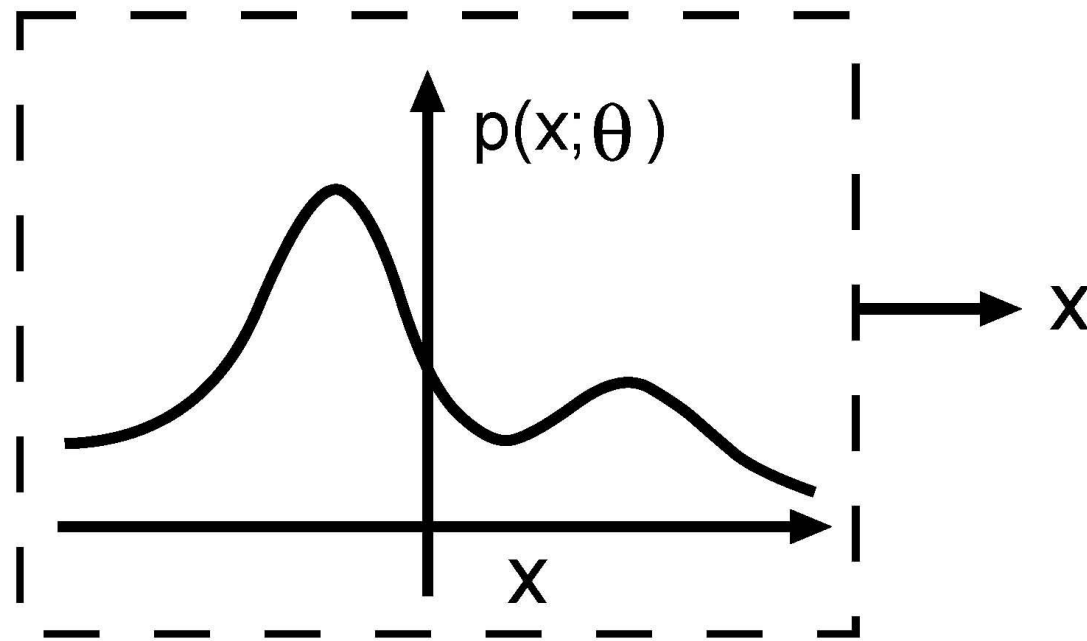
- B) **Bayesian estimation: θ is random**
 - **Minimum mean squared error (MMSE)**

Bayesian estimation

- Classical and Bayesian estimation
- Linear minimum mean square-error (MMSE) estimation
- Orthogonality principle
- Yule-Walker equation

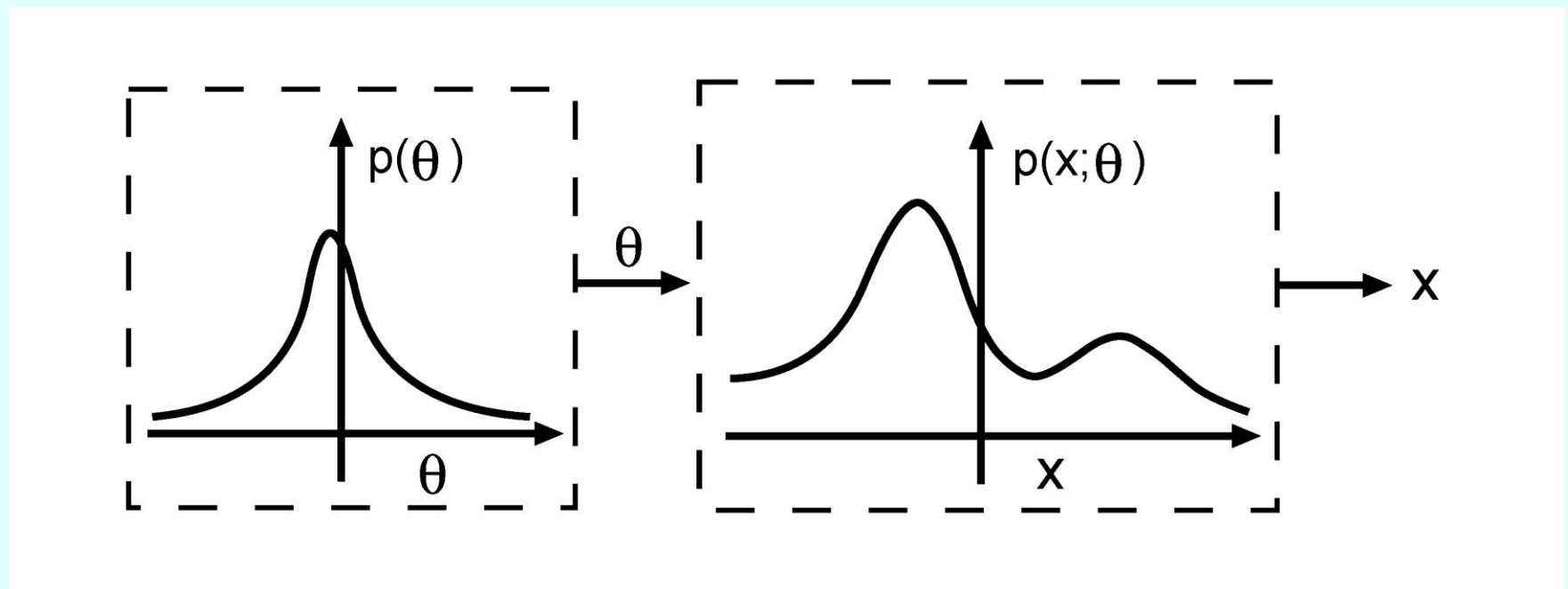
1. Classical and Bayesian estimation

In classical estimation, the unknown parameter θ is viewed as ***nonrandom***. Statistical model is specified entirely by conditional probability $p(x, \theta)$.

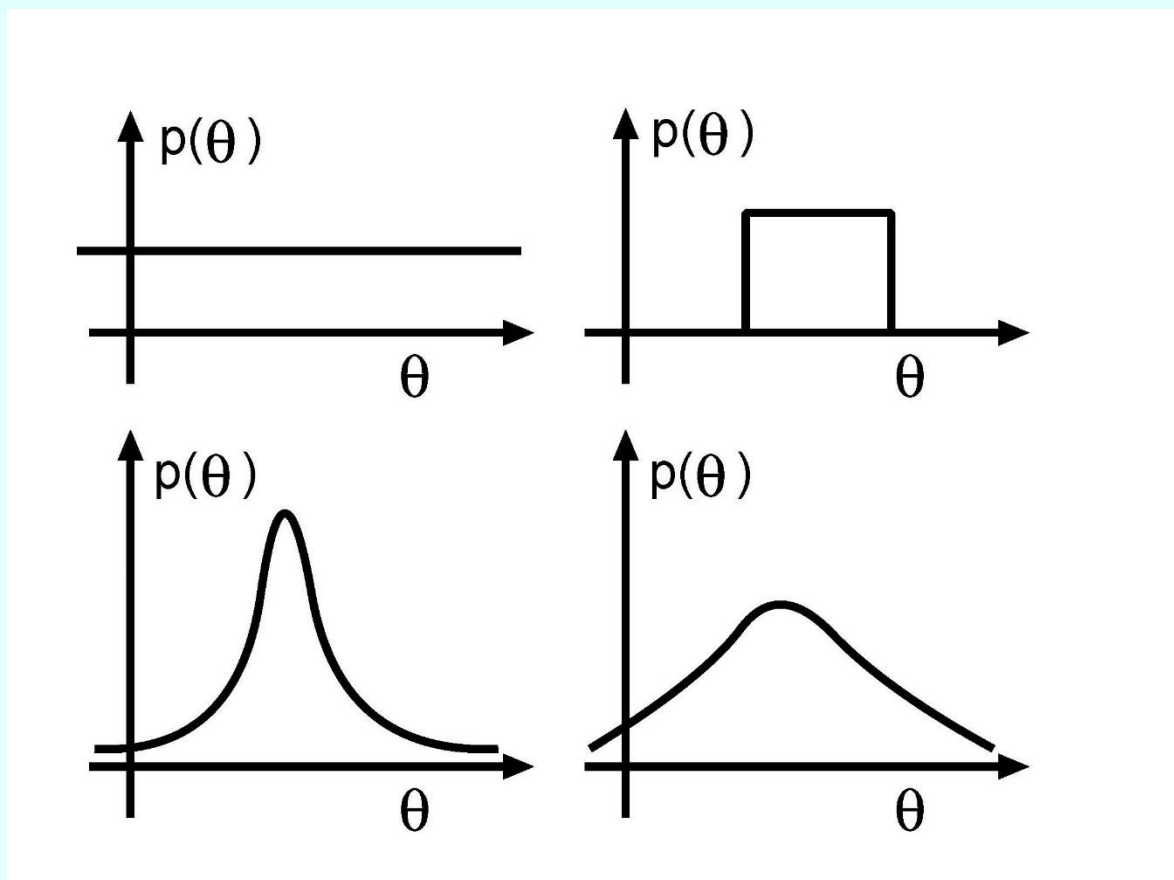


1. Classical and Bayesian estimation

In Bayesian estimation, the unknown parameter θ is viewed as **random**. Statistical model is specified in terms of conditional probability $p(x|\theta)$ and a prior distribution $p(\theta)$ on θ .



The prior $p(\theta)$ can be used as a “prior knowledge.”



- By Bayes' rule, we may express the posterior distribution of θ given x as

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)} = \frac{p(x|\theta)p(\theta)}{\int p(x|\theta)p(\theta)d\theta}$$

- Definition of Mean Squared Error $E\{(\theta - \hat{\theta})^2\}$
- Classical (θ is nonrandom):

$$MSE(\hat{\theta}) = \int (\hat{\theta} - \theta)^2 p(x, \theta) dx$$

- Bayesian (θ is random):

$$Bmse(\hat{\theta}) = \iint (\theta - \hat{\theta})^2 p(x, \theta) dx d\theta$$

■ Minimum Mean Squared Error (MMSE)

From $p(x, \theta) = p(\theta|x)p(x)$,

$$Bmse(\hat{\theta}) = \int \left[\int (\theta - \hat{\theta})^2 p(\theta|x)p(x)d\theta \right] dx$$

To minimize the inside of $[\cdot]$,

$$\begin{aligned} \frac{\partial}{\partial \hat{\theta}} \int (\theta - \hat{\theta})^2 p(\theta|x)d\theta &= \int \frac{\partial}{\partial \hat{\theta}} (\theta - \hat{\theta})^2 p(\theta|x)d\theta \\ &= \int -2(\theta - \hat{\theta})p(\theta|x)d\theta \\ &= -2 \int \theta p(\theta|x)d\theta + 2\hat{\theta} \int p(\theta|x)d\theta = 0 \end{aligned}$$

Hence, $\hat{\theta} = \int \theta p(\theta|x)d\theta = E(\theta|x)$.

The best estimation is given by expectation value of θ under condition that x is observed.

Exercise

Imagine we want to estimate an unknown voltage parameter θ .

- **Prior:** We have a prior belief that θ is uniformly distributed between 0 and 2 Volts.

$$p(\theta) = \frac{1}{2}, 0 \leq \theta \leq 2$$

- **Likelihood:** We measure the voltage with a noisy sensor. The observed measurement x is uniformly distributed between 0 and the true voltage θ .

$$p(x|\theta) = \frac{1}{\theta}, 0 \leq x \leq \theta$$

Question: You observe a specific measurement $x = 1$. Find the optimal Bayesian MMSE estimate $\hat{\theta}$.

2. Linear MMSE Estimation

With respect to linear estimator

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n X[n] + a_N$$

Minimize Mean Squared Error

$$Bmse(\hat{\theta}) = E_{X,\theta} [(\theta - \hat{\theta})^2]$$

$$\begin{aligned} \frac{\partial Bmse(\hat{\theta})}{\partial a_N} &= \frac{\partial}{\partial a_N} E[(\theta - \sum_{n=0}^{N-1} a_n X[n] - a_N)^2] \\ &= -2E[\theta - \sum_{n=0}^{N-1} a_n X[n] - a_N] = 0 \end{aligned}$$

Substituting $a_N = E[\theta] - \sum_{n=0}^{N-1} a_n E[X[n]]$,

$$Bmse(\hat{\theta})$$

$$= E[\{\sum_{n=0}^{N-1} a_n (X[n] - E[X[n]]) - (\theta - E[\theta])\}^2]$$

$$= E[\{\mathbf{a}^T (\mathbf{X} - E[\mathbf{X}]) - (\theta - E[\theta])\}^2]$$

$$= E[\mathbf{a}^T (\mathbf{X} - E[\mathbf{X}]) (\mathbf{X} - E[\mathbf{X}])^T \mathbf{a}]$$

$$- E[\mathbf{a}^T (\mathbf{X} - E[\mathbf{X}]) (\theta - E[\theta])]$$

$$- E[(\theta - E[\theta]) (\mathbf{X} - E[\mathbf{X}])^T \mathbf{a}]$$

$$+ E[(\theta - E[\theta])^2]$$

$$= \mathbf{a}^T E[(\mathbf{X} - E[\mathbf{X}]) (\mathbf{X} - E[\mathbf{X}])^T] \mathbf{a}$$

$$- \mathbf{a}^T E[(\mathbf{X} - E[\mathbf{X}]) (\theta - E[\theta])]$$

$$- E[(\theta - E[\theta]) (\mathbf{X} - E[\mathbf{X}])^T] \mathbf{a}$$

$$+ E[(\theta - E[\theta])^2]$$

$$= \mathbf{a}^T \mathbf{C}_{XX} \mathbf{a} - \mathbf{a}^T \mathbf{C}_{X\theta} - \mathbf{C}_{\theta X} \mathbf{a} + \mathbf{C}_{\theta\theta}$$

Here,

$$\mathbf{X} = [X[0], X[1], \dots, X[N-1]]^T,$$

$$\mathbf{a} = [a_0, a_1, \dots, a_{N-1}]^T,$$

covariance matrices are

$$\mathbf{C}_{XX} = E\{(\mathbf{X} - E(\mathbf{X}))(\mathbf{X} - E(\mathbf{X}))^T\} \in \Re^{N \times N},$$

$$\mathbf{C}_{\theta\theta} = E\{(\theta - E(\theta))(\theta - E(\theta))^T\} \in \Re^{1 \times 1}$$

$$\mathbf{C}_{\theta X} = E\{(\theta - E(\theta))(\mathbf{X} - E(\mathbf{X}))^T\} = \mathbf{C}_{X\theta}^T \in \Re^{1 \times N}$$

$$\frac{\partial Bmse}{\partial \mathbf{a}} = 2\mathbf{C}_{XX}\mathbf{a} - 2\mathbf{C}_{X\theta} = 0$$

Hence,

$$\mathbf{a} = \mathbf{C}_{XX}^{-1} \mathbf{C}_{X\theta}$$

$$\begin{aligned}\hat{\theta} &= \mathbf{a}^T \mathbf{X} + a_N \\&= \mathbf{C}_{X\theta}^T \mathbf{C}_{XX}^{-1} \mathbf{X} + E(\theta) - \mathbf{C}_{X\theta}^T \mathbf{C}_{XX}^{-1} E(\mathbf{X}) \\&= E(\theta) + \mathbf{C}_{\theta X} \mathbf{C}_{XX}^{-1} (\mathbf{X} - E(\mathbf{X})) \\&\quad \text{(in case of } E(\mathbf{X}) = 0, E(\theta) = 0) \\&= \mathbf{C}_{\theta X} \mathbf{C}_{XX}^{-1} \mathbf{X}\end{aligned}$$

Corresponding minimum mean square error is

$$\begin{aligned} Bmse(\hat{\theta}) &= \mathbf{a}^T \mathbf{C}_{XX} \mathbf{a} - \mathbf{a}^T \mathbf{C}_{X\theta} - \mathbf{C}_{\theta X} \mathbf{a} + \mathbf{C}_{\theta\theta} \\ &= \mathbf{C}_{X\theta}^T \mathbf{C}_{XX}^{-1} \mathbf{C}_{XX} \mathbf{C}_{XX}^{-1} \mathbf{C}_{X\theta} - \mathbf{C}_{X\theta}^T \mathbf{C}_{XX}^{-1} \mathbf{C}_{X\theta} \\ &\quad - \mathbf{C}_{\theta X} \mathbf{C}_{XX}^{-1} \mathbf{C}_{X\theta} + \mathbf{C}_{\theta\theta} \\ &= \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta X} \mathbf{C}_{XX}^{-1} \mathbf{C}_{X\theta}. \end{aligned}$$

LMMSE Properties

- **Unbiased on Average:** Taking the expectation of both sides reveals $E[\hat{\theta}] = E[\theta]$. The linear estimator is perfectly centered.
- **Fluctuation Modeling:** The optimal linear estimator starts with the prior mean of the parameter ($E[\theta]$) as a baseline, and then modifies its guess up or down based on how much the observed data deviates from its expected value ($X[n] - E[X[n]]$).

Exercise

Consider the following random variable,

$$X[n] = \theta + W[n], \quad (n = 0, 1, \dots, N - 1)$$

$W[n]$ is independent Gaussian with mean 0 and variance σ^2 , DC component θ obeys uniform distribution $U(-\theta_0, \theta_0)$. θ and $W[n]$ are independent. Find the minimum mean square error estimation for θ .

Answer

From $E(\theta) = 0, E(X[n]) = 0$ (Namely, $E(\mathbf{X}) = 0$).

$$\begin{aligned}\mathbf{C}_{XX} &= E(\mathbf{X}\mathbf{X}^T) = E[(\theta\mathbf{1} + \mathbf{W})(\theta\mathbf{1} + \mathbf{W})^T] \\ &= E(\theta^2)\mathbf{1}\mathbf{1}^T + \sigma^2\mathbf{I}\end{aligned}$$

$$\begin{aligned}\mathbf{C}_{\theta X} &= E(\theta\mathbf{X}^T) = E[\theta(\theta\mathbf{1} + \mathbf{W})^T] = E(\theta^2)\mathbf{1}^T \\ &\text{where } \mathbf{1} = (1, 1, \dots, 1)^T\end{aligned}$$

Therefore ,

$$\begin{aligned}\hat{\theta} &= \mathbf{C}_{\theta X} \mathbf{C}_{XX}^{-1} \mathbf{X} \\ &= \sigma_{\theta}^2 \mathbf{1}^T (\sigma_{\theta}^2 \mathbf{1} \mathbf{1}^T + \sigma^2 \mathbf{I})^{-1} \mathbf{X} \quad (\leftarrow \sigma_{\theta}^2 = E(\theta^2)) \\ &= \frac{\sigma_{\theta}^2}{\sigma_{\theta}^2 + \sigma^2 / N} \bar{\mathbf{X}} \quad (\text{From Woodbury's identity})\end{aligned}$$

$$\begin{aligned}\left(\mathbf{I} + \frac{\sigma_{\theta}^2}{\sigma^2} \mathbf{1} \mathbf{1}^T \right)^{-1} &= \mathbf{I} - \frac{\frac{\sigma_{\theta}^2}{\sigma^2} \mathbf{1} \mathbf{1}^T}{1 + N + \frac{\sigma_{\theta}^2}{\sigma^2}} \\ &= \frac{\frac{\theta_0^2}{3}}{\frac{\theta_0^2}{3} + \sigma^2 / N} \bar{\mathbf{X}} \quad (\leftarrow \sigma_{\theta}^2 = \frac{\theta_0^2}{3})\end{aligned}$$

3. Orthogonality Principle

Problem: With respect to linear estimator $\hat{\theta} = \sum_{n=0}^{N-1} a_n X[n]$, error is given by

$$\epsilon = \theta - \hat{\theta} = \theta - \sum_{n=0}^{N-1} a_n X[n].$$

Assuming $E(X) = 0, E(\theta) = 0$, then $a_N = 0$.

The minimum mean square error is

$$Bmse(\hat{\theta}) = E\{\epsilon^2\} = E\{(\theta - \sum_{n=0}^{N-1} a_n X[n])^2\}$$

Orthogonality Principle

If coefficients a_n are selected in such a way that error ϵ is orthogonal to random variables $X[0], X[1], \dots, X[N-1]$, i.e.

$$E\{\epsilon X[n]\} = E\left\{\left(\theta - \sum_{i=0}^{N-1} a_i X[i]\right) X[n]\right\} = 0$$
$$(n = 0, 1, \dots, N-1),$$

the mean square error $Bmse$ is minimal.

[Proof] Local minimum condition of $Bmse$ (with respect to a_n) is

$$\frac{\partial Bmse}{\partial a_n} = E\left\{2\left(\theta - \sum_{i=0}^{N-1} a_i X[i]\right)(-X[n])\right\} = 0$$
$$(n = 0, 1, \dots, N-1).$$

Determination of a by Orthogonality Principle

Orthogonality principle

$$E\left\{\left(\theta - \sum_{j=0}^{N-1} a_j X[j]\right)X[n]\right\} = 0$$

can be decomposed as

$$E\{\theta X[n]\} = \sum_{j=0}^{N-1} a_j E\{X[j]X[n]\}$$

Denoting $[C_{XX}]_{jn} = E\{X[j]X[n]\}$, $[C_{\theta X}]_n = E\{\theta X[n]\}$,

$$[C_{\theta X}]_0 = [C_{XX}]_{00}a_0 + [C_{XX}]_{01}a_1 + \cdots + [C_{XX}]_{0N-1}a_{N-1}$$
$$[C_{\theta X}]_1 = [C_{XX}]_{10}a_0 + [C_{XX}]_{11}a_1 + \cdots + [C_{XX}]_{1N-1}a_{N-1}$$

$\vdots$

$$[C_{\theta X}]_{N-1} = [C_{XX}]_{N-10}a_0 + [C_{XX}]_{N-11}a_1 + \cdots + [C_{XX}]_{N-1N-1}a_{N-1}$$



These are called Yule-Walker equation.

If $X[0], X[1], \dots, X[N - 1]$ are linearly independent, inverse matrix $\mathbf{C}_{XX}^{-1}$ exists (if $\mathbf{C}_{XX}$ is invertible), $\mathbf{a}$ is obtained as,

$$\mathbf{a} = \mathbf{C}_{XX}^{-1} \mathbf{C}_{\theta X}$$

Exercise

Suppose you are trying to estimate a zero-mean random signal value, $X[n]$, based on its two immediate past observations, $X[n - 1]$ and $X[n - 2]$. The linear estimator is defined as:

$$\hat{X}[n] = a_1 X[n - 1] + a_2 X[n - 2]$$

The random process is stationary with a known autocorrelation function, $R_{XX}[k] = E\{X[n]X[n - k]\}$. Specifically, you are given the following correlation properties:

- $R_{XX}[0] = 4$ (The variance of the signal)
 - $R_{XX}[1] = 2$ (The correlation with a lag of 1 sample)
 - $R_{XX}[2] = 1$ (The correlation with a lag of 2 samples)
- a) Use the Orthogonality Principle to set up the system of equations for the optimal coefficients a_1 and a_2 .
 - b) Arrange these into the Yule-Walker Matrix Form.
 - c) Solve for the optimal filter coefficients a_1 and a_2 .

Mean Square Error

From orthogonality principle

$$E\{\epsilon X[n]\} = E\{(\theta - \hat{\theta})X[n]\} = 0,$$

for arbitrary c_n ($n = 0, 1, \dots, N - 1$)

$$E\{(\theta - \hat{\theta})(c_0X[0] + c_1X[1] + \dots + c_{N-1}X[N - 1])\} = 0$$

Substituting $c_i = a_i$,

$$E\{(\theta - \hat{\theta})\hat{\theta}\} = 0.$$

Therefore,

$$\begin{aligned} Bmse(\hat{\theta}) &= E\{|\theta - \hat{\theta}|^2\} = E\{(\theta - \hat{\theta})(\theta - \hat{\theta})\} \\ &= E\{\theta\theta - \theta\hat{\theta} - \hat{\theta}\theta + \hat{\theta}\hat{\theta}\} \\ &= E\{\theta\theta - \hat{\theta}\theta - (\theta - \hat{\theta})\hat{\theta}\} \\ &= E\{\theta\theta - \hat{\theta}\theta\} \\ &= E\{(\theta - \hat{\theta})\theta\} = E\{\theta\theta\} - E\{\sum_{n=0}^{N-1} a_n X[n] \theta\} \\ &= E\{\theta\theta\} - \sum_{n=0}^{N-1} a_n E\{X[n]\theta\} \\ &= C_{\theta\theta} - \\ &\quad (a_0[\mathbf{C}_{X\theta}]_0 + a_1[\mathbf{C}_{X\theta}]_1 + \cdots + a_{N-1}[\mathbf{C}_{X\theta}]_{N-1}) \\ &= C_{\theta\theta} - \mathbf{a}\mathbf{C}_{X\theta} = \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta X}\mathbf{C}_{XX}^{-1}\mathbf{C}_{X\theta} \end{aligned}$$